

FLUOR

VERTICAL VESSEL FOUNDATIONS

PURPOSE

This document establishes guidelines and recommended procedures for the design of vertical vessel foundations.

SCOPE

This document includes the following major sections:

- GENERAL
- DESIGN CONDITIONS
- ANCHOR BOLTS
- PEDESTAL DESIGN
- FOOTING DESIGN
- REFERENCES
- ATTACHMENTS

APPLICATION

This document applies to all types of foundations for vertical vessels and self supported stacks.

GENERAL

Before designing any vertical vessel foundations, the Lead Structural Engineer will meet with the Lead Vessel Engineer to agree on consistent wind and seismic loadings, corrosion allowances for anchor bolts, anchor bolt types, and any special requirements dictated by the client.

At the time of the foundation drawing squad check by the Lead Structural Engineer, the vessel drawing should be compared to the data used on the design calculations to ensure that no major platform changes, anchor bolt changes, or vessel size changes have occurred.

Computer programs are available for the design of vertical vessel foundations, and should be used whenever possible to expedite foundation design.

DESIGN CONDITIONS

Vertical Loads

Erection Weight: Fabricated weight of vessel, generally taken from vessel drawing, plus internals, platforms, manways, and piping that are actually erected with the vessel.

Empty Weight: Fabricated weight of vessel plus weight of internals, piping, insulation, fireproofing, and platforms; generally taken from vessel drawing.

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Operating Weight: Empty weight plus weight of operating liquid or catalyst; generally taken from vessel drawing.

Test Weight: Empty weight plus weight of water required for hydrostatic test; generally taken from vessel drawing.

It should be determined whether a hydrostatic test will actually be done in the field. It is generally desirable to design for test weight since unforeseen circumstances may occur.

The above loads should be considered as dead loads when applying load factors used in ultimate strength design.

Wind Loads

Wind loads should be calculated in accordance with the job specifications and Structural Engineering Guideline 000.215.1215: Wind Load Calculation.

Wind loads are normally calculated by the Mechanical group, however, verification by the Structural group may be required. Wind loads calculated by hand should be compared to the computer printout. If the two results compare favorably, the mechanical results should be used for foundation design. If the results do not compare favorably, the differences shall be jointly discussed and resolved.

When calculating or checking wind loads, due consideration should be given to factors which may significantly affect total wind loads such as the application of dynamic gust factors or the presence of spoilers on the vessel.

Seismic Loads

In earthquake zones, seismic forces determined in accordance with job specifications, usually conforming with the UBC (Uniform Building Code), will replace wind forces when greater. Refer to Structural Engineering Guideline 000.215.1216: Earthquake Engineering, for seismic design considerations and procedures.

Seismic loads calculated by the Mechanical group shall be independently verified as appropriate by the Structural group.

Thermal Loads

Thrusts due to thermal expansion of piping will be included in the operating load combinations, when deemed advisable. Dead load factors will be applied to the resultants of piping thermal loadings. Consult with the Pipe Stress Engineer for any thermal loads that are to be considered.

Load Combinations

Load Combinations shall be calculated in accordance with Structural Engineering Specification 000.215.00910, Structural Engineering Criteria.

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The allowable soil bearing pressure will not be increased for any combination, unless dictated by job specifications.

ANCHOR BOLTS

Design procedures for sizing anchor bolts, factoring anchor bolt loads, determining pretensioning, and checking embedment are covered in Structural Engineering Guideline 000.215.1207: Anchor Bolt Design Criteria.

Sizes shall be checked by the Structural Engineer in accordance with following formula:

$$\text{Tension} = \frac{4 M_u}{N \times BC} - \frac{W_u}{N} \quad (\text{Equation 1})$$

where

M_u = Factored overturning moment at base of vessel

N = Number of anchor bolts (vessel drawing)

BC = Diameter of bolt circle (vessel drawing)

W_u = Factored weight of vessel

Each load case shall be considered. In non-seismic areas, an empty vessel combined with wind will usually control. In seismic areas, the full weight will produce the maximum moment. Seismic shear may govern the design for short vessels.

PEDESTAL DESIGN**Sizing**

Concrete pedestals supporting vertical vessels will be sized with pedestal faces in even inch increments according to the following criteria:

- Face-to-face pedestal size will be no less than the larger of:

Bolt circle + 8 inches

Bolt circle + 8 bolt diameters

Bolt circle + sleeve diameter + 6 inches

(Equation 2)

- Pedestals 4'- 0" and larger will be octagonal in shape; smaller than 4'- 0", use a square shape.

It is normally desirable to make the pedestal deep enough to contain the anchor bolts and keep them out of the mat.

Reinforcing

The pedestal will be tied to the footing with sufficient dowels around the pedestal

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perimeter to prevent separation of the pedestal and footing.

Dowels

Dowels are customarily sized by computing the maximum tension existing at the pedestal perimeter due to overturning moments (using allowable stress load combinations) in accordance with the following formula:

For seismic cases

$$Tension, (F_u) = \left[1.0 \times \frac{4 M_{ped}}{N \times BC} \right] - \left[0.9 \times \frac{W \text{ (or } W_o)}{N} \right] \quad (\text{Equation 3a})$$

For wind cases,

$$Tension, (F_u) = \left[1.3 \times \frac{4 M_{ped}}{N \times BC} \right] - \left[0.9 \times \frac{W}{N} \right] \quad (\text{Equation 3b})$$

$$Area \text{ Required} = \frac{Tension}{Design \text{ Stress}} = \frac{F_u}{\phi f_y} = \frac{F_u}{0.9 f_y} \quad (\text{Equation 4})$$

where

- M_{ped} = Maximum overturning moment at base of pedestal
- N = Number of bars (assumed)
- BC = Pedestal size minus 6 inches
- W = Empty weight of vessel and pedestal
- F_u = Maximum ultimate tension in reinforcing bar
- W_o = Operating weight of vessel and pedestal

Minimum Pedestal Reinforcement

Octagons 4'- 0" to 5'- 10": 8 - #4 vertical with #4 ties at 15 inches maximum.

Octagons 6'- 0" to 8'- 10": 16 - #4 vertical with #4 ties at 15 inches maximum.

Octagons 9'- 0" to 12'- 0": 24 - #5 vertical with #5 ties at 15 inches maximum.

Octagons larger than 12'- 0": 32 - #5 vertical with #5 ties at 15 inches maximum.

Pedestals larger than 8'- 0" in diameter will have a mat of reinforcing steel at the top.

The requirement for additional vertical reinforcing for anchor bolt development will be checked in accordance with Structural Engineering Guideline 000.215.1207.

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The following procedures cover footing design for soil supported foundations. For foundations supported on drilled shafts or driven piles, refer to Structural Engineering Guideline 000.215.1231: Drilled Pile Foundations, and Structural Engineering Guideline 000.215.1232: Driven Pile Foundations.

Sizing

Footings for vertical vessels will normally be octagonal in shape and sized with footing faces in even inch increments to simplify formwork. Offsets and combined footings will be avoided where possible. Footings smaller than 7'- 0" in diameter will be square.

The footing thickness will be 12 inches minimum and thickened in 3-inch increments. The footing thickness required for adequate embedment of pedestal reinforcement should be checked in accordance with ACI 318, Chapter 12.

The footing thickness shall also be checked for top tension without top reinforcing in accordance with ACI 318, Chapter 22. If not adequate, either a thicker footing or top reinforcing shall be used. The curves in Attachment 03 may be used as appropriate. Note that the load factor of 1.4 in Attachment 03 is based on a UBC specified combination of dead and seismic load. For non-seismic conditions, a separate calculation using a lower load factor would be appropriate.

For the first trial, the diameter of an octagonal footing may be approximated by the following formula:

$$\text{Diameter} = \sqrt[3]{\frac{M_{ftg}}{SB}} \quad (\text{Equation 5})$$

where

M_{ftg} = Overturning moment at the base of the footing in foot kips

SB = Allowable gross soil bearing in ksf

Soil Bearing

Soil bearing pressure will be checked for maximum allowable on the diagonal.

Soil bearing pressure used for footing design will be computed on the flat.

Soil bearing pressure is computed based on the ratio of load eccentricity to footing size (e/D). When total footing area is not in compression

$$e / D > 0.123,$$

the soil bearing pressure will be computed using the charts in Attachment 01 where

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$$SB = \frac{L P}{A} \quad (\text{Equation 6})$$

When total octagon footing area is in compression

$$e / D \leq 0.123,$$

the soil bearing pressure will be computed using the combined stresses formula:

$$SB = \frac{P}{A} \pm \frac{M_{fig}}{Z}$$

$$SB \text{ diagonal} = \frac{P}{A} \left(1 \pm \frac{8.19 e}{D} \right) \quad (\text{Equation 7a})$$

$$SB \text{ flat} = \frac{P}{A} \left(1 \pm \frac{7.57 e}{D} \right) \quad (\text{Equation 7b})$$

Stability Ratio

Minimum stability ratio should be 1.5 for wind or seismic and 1.0 for dead load unless job specifications require greater stability. Normally, stability ratio is not checked for thermal cases.

Reinforcing

Consider the critical sections for moment and shear taken with respect to the face of a square of area equivalent to that of the pedestal.

Moment should be checked at the face of the equivalent square.

Shear, as a measure of diagonal tension, should be checked at the face of the equivalent square. If excessive, check using the critical section specified in ACI 318-99, Section 11.1.3.1.

The moment and shear shall be figured for a unit width strip as a simple cantilever from the equivalent square. The resulting reinforcing steel shall be placed continuously across the entire footing in a grid pattern.

If tensile stress in the upper face of footing exceeds the allowable concrete stress, top steel shall be used, provided it is not feasible to increase the footing thickness.

The minimum bottom footing reinforcing will be #5 bars at 12 inches OC (On Center) each way.

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REFERENCES

American Concrete Institute. Building Code Requirements for Structural Concrete. ACI 318-99. 1999

International Conference of Building Officials, Uniform Building Code. UBC 97. 1997

Structural Engineering
Specification 000.215.00910: Structural Design Criteria

Structural Engineering
Guideline 000.215.1207: Anchor Bolt Design Criteria

Structural Engineering
Guideline 000.215.1215: Wind Load Calculation

Structural Engineering
Guideline 000.215.1216: Earthquake Engineering

Structural Engineering
Guideline 000.215.1231: Drilled Pile Foundations

Structural Engineering
Guideline 000.215.1232: Driven Pile Foundations

ATTACHMENTS

Attachment 01: (31Mar05)
Foundation Pressures For Octagon Bases

Attachment 02: (31Mar05)
Octagon Properties

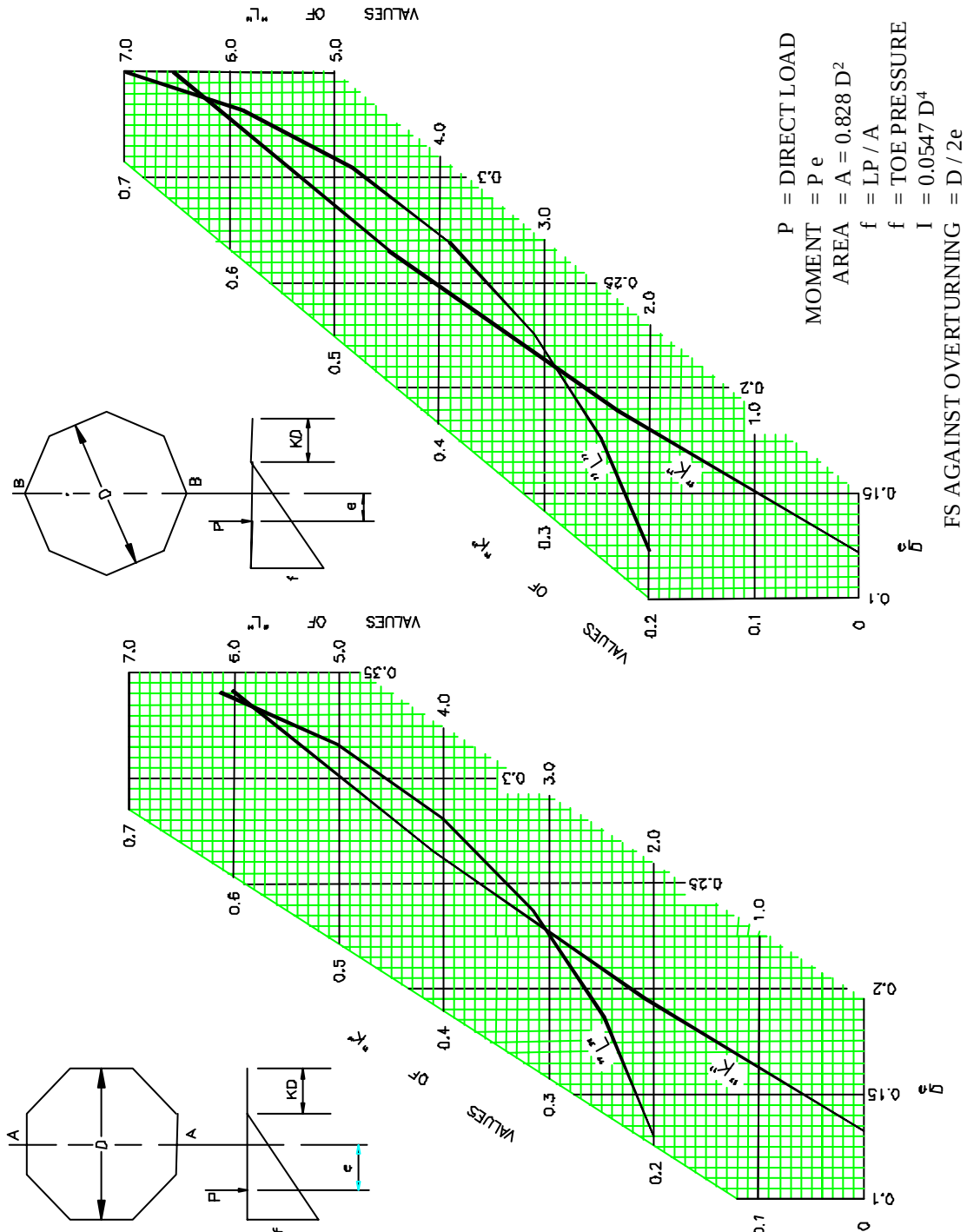
Attachment 03: (31Mar05)
Footing Thickness Required Using No Top Steel

Attachment 04: (31Mar05)
Sample Design 1: Vertical Vessel Foundation

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VERTICAL VESSEL FOUNDATIONS

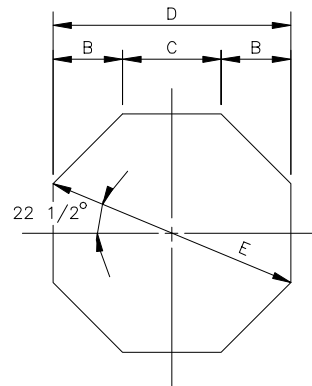
Foundation Pressures For Octagon Bases



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Octagon Properties



$$A = \text{AREA (SF)} = 0.8284272 D^2$$

$$B = C \times \sin 45^\circ = 0.2928932 D$$

$$C = \text{LENGTH OF SIDE} = 0.4142136 D$$

$$E = \text{LENGTH OF DIAMETER} = 1.0823922 D$$

$$Z_e = \text{SEC MOD DIAMETER} = 0.1011422 D^3$$

$$Z_d = \text{SEC MOD FLAT} = Z_e E / D$$

$$I = \text{MOMENT OF INERTIA} = Z_e E / 2$$

D		C		B	A	Z	E
FT - IN.	FT	FT - IN.		FT - IN.	FT ²	FT ³	FT
4- 0-1/4	4.024	1- 8		1- 2-1/8	13.41	6.59	4.36
4- 2-3/4	4.225	1- 9		1- 2-7/8	14.78	7.63	4.58
4- 5-1/8	4.426	1- 10		1- 3-9/16	16.23	8.77	4.79
4- 7-1/2	4.627	1- 11		1- 4-1/4	17.73	10.02	5.01
4- 10	4.828	2- 0		1- 5	19.31	11.38	5.23
5- 0-3/8	5.030	2- 1		1- 5-11/16	20.96	12.87	5.44
5- 2-3/4	5.231	2- 2		1- 6-3/8	22.67	14.48	5.66
5- 5-1/4	5.432	2- 3		1- 7-1/8	24.44	16.21	5.88
5- 7-5/8	5.633	2- 4		1- 7-13/16	26.29	18.08	6.10
5- 10	5.834	2- 5		1- 8-1/2	28.20	20.08	6.32
6- 0-1/2	6.036	2- 6		1- 9-1/4	30.18	22.24	6.53
6- 2-7/8	6.237	2- 7		1- 9-15/16	32.23	24.54	6.75
6- 5-1/4	6.438	2- 8		1- 10-5/8	34.34	26.99	6.97
6- 7-5/8	6.639	2- 9		1- 11-5/8	36.51	29.60	7.19
6- 10-1/8	6.840	2- 10		2- 0-1/16	38.76	32.37	7.40
7- 0-1/2	7.042	2- 11		2- 0-3/4	41.08	35.32	7.62
7- 2-7/8	7.243	3- 0		2- 1-7/16	43.46	38.43	7.84
7- 5-3/8	7.444	3- 1		2- 2-3/16	45.91	41.72	8.06
7- 7-3/4	7.645	3- 2		2- 2-7/8	48.42	45.19	8.27
7- 10-1/8	7.846	3- 3		2- 3-9/16	51.00	48.85	8.49
8- 0-5/8	8.047	3- 4		2- 4-5/16	53.64	52.70	8.71
8- 3	8.249	3- 5		2- 5	56.37	56.77	8.93
8- 5-3/8	8.450	3- 6		2- 5-11/16	59.15	61.02	9.15
8- 7-7/8	8.651	3- 7		2- 6-7/16	62.00	65.48	9.36
8- 10-1/4	8.852	3- 8		2- 7-1/8	64.91	70.15	9.58
9- 0-5/8	9.053	3- 9		2- 7-13/16	67.90	75.04	9.79
9- 3	9.250	3- 10		2- 8-1/2	70.88	80.05	10.01
9- 5-1/2	9.456	3- 11		2- 9-1/4	74.07	85.52	10.23
9- 7-7/8	9.657	4- 0		2- 9-5/16	77.26	91.09	10.45
9- 10-3/8	9.858	4- 1		2- 10-11/16	80.51	96.89	10.67
10- 0-3/4	10.059	4- 2		2- 11-3/8	83.82	102.94	10.89
10- 3-1/8	10.260	4- 3		3- 0-1/16	87.21	109.24	11.10

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Octagon Properties

D		C		B	A	Z	E
FT	— IN.	FT	FT — IN.	FT — IN.	FT ²	FT ³	FT
10—	5—1/2	10.462	4— 4	3— 0—3/4	90.67	115.82	11.32
10—	8	10.663	4— 5	3— 1—1/2	94.19	122.62	11.54
10—	10—3/8	10.864	4— 6	3— 2—3/16	97.78	129.69	11.76
11—	0—3/4	11.065	4— 7	3— 2—7/8	101.43	137.02	11.97
11—	3—1/4	11.266	4— 8	3— 3—5/8	105.15	144.62	12.19
11—	5—5/8	11.468	4— 9	3— 4—15/16	108.95	152.54	12.41
11—	8	11.669	4— 10	3— 5	112.80	160.71	12.62
11—	10—3/8	11.870	4— 11	3— 5—11/16	116.72	169.15	12.84
12—	0—7/8	12.071	5— 0	3— 6—7/16	120.71	177.89	13.06
12—	3—1/4	12.272	5— 1	3— 7—1/8	124.76	186.93	13.23
12—	5—5/8	12.474	5— 2	3— 7—13/16	128.90	196.31	13.49
12—	8—1/8	12.675	5— 3	3— 8—9/16	133.09	205.96	13.72
12—	10—1/2	12.876	5— 4	3— 9—1/4	137.35	215.91	13.93
13—	0—7/8	13.077	5— 5	3— 9—15/16	141.67	226.18	14.15
13—	3—3/8	13.278	5— 6	3— 10—11/16	146.06	236.77	14.37
13—	5—3/4	13.479	5— 7	3— 11—3/8	150.51	247.69	14.58
13—	8—1/8	13.681	5— 8	4— 0—1/16	155.06	258.99	14.80
13—	10—5/8	13.882	5— 9	4— 0—13/16	159.65	270.58	15.02
14—	1	14.083	5— 10	4— 1—1/2	164.30	282.50	15.24
14—	3—3/8	14.284	5— 11	4— 2—3/16	169.03	294.77	15.45
14—	5—7/8	14.485	6— 0	4— 2—15/16	173.62	307.39	15.68
14—	8—1/4	14.686	6— 1	4— 3—5/8	178.67	320.36	15.90
14—	10—5/8	14.888	6— 2	4— 4—5/16	183.62	333.77	16.11
15—	1—1/8	15.088	6— 3	4— 5—1/16	188.59	347.40	16.33
15—	3—1/2	15.290	6— 4	4— 5—3/4	193.67	361.54	16.55
15—	5—7/8	15.491	6— 5	4— 6—7/16	198.80	375.98	16.77
15—	8—1/4	15.692	6— 6	4— 7—1/8	203.99	390.81	16.98
15—	10—3/4	15.893	6— 7	4— 7—7/8	209.25	406.02	17.20
16—	1—1/8	16.095	6— 8	4— 8—9/16	214.60	421.70	17.42
16—	3—5/8	16.296	6— 9	4— 9—5/16	220.00	437.70	17.65
16—	6	16.497	6— 10	4— 10	225.46	454.10	17.85
16—	8—3/8	16.698	6— 11	4— 10—11/16	230.98	470.90	18.07
16—	10—3/4	16.899	7— 0	4— 11—3/8	236.58	488.11	18.23
17—	1—1/4	17.101	7— 1	5— 0—1/8	242.27	505.82	18.51
17—	3—5/8	17.302	7— 2	5— 0—13/16	248.00	523.87	18.72
17—	6	17.503	7— 3	5— 1—1/2	253.79	542.34	18.94
17—	8—1/2	17.704	7— 4	5— 2—1/4	259.66	561.24	19.16
17—	10—7/8	17.905	7— 5	5— 2—15/16	265.58	580.57	19.38
18—	1—1/4	18.107	7— 6	5— 3—5/8	271.61	600.44	19.60
18—	3—3/4	18.308	7— 7	5— 4—3/8	277.67	620.66	19.81
18—	6—1/8	18.509	7— 8	5— 5—1/16	283.81	641.33	20.03
18—	8—1/2	18.710	7— 9	5— 5—3/4	290.00	662.45	20.25
18—	10—7/8	18.911	7— 10	5— 6—7/16	296.27	683.03	20.47
19—	1—3/8	19.113	7— 11	5— 7—3/16	302.64	706.19	20.69

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Octagon Properties

D		C		B	A	Z	E
FT - IN.	FT	FT - IN.	FT	FT - IN.	FT ²	FT ³	FT
19- 3-3/4	19.313	8- 0		5- 7-7/8	309.00	728.59	20.90
19- 6-1/8	19.515	8- 1		5- 8-9/16	315.50	751.69	21.12
19- 8-5/8	19.716	8- 2		5- 9-5/16	322.03	775.16	21.34
19- 11	19.917	8- 3		5- 10	328.63	799.11	21.55
20- 1-3/8	20.118	8- 4		5- 10-11/16	335.30	823.54	21.78
20- 3-7/8	20.320	8- 5		5- 11-7/16	342.06	848.60	21.99
20- 6-1/4	20.521	8- 6		6- 0-1/8	348.86	874.03	22.21
20- 8-3/4	20.722	8- 7		6- 0-7/8	355.72	899.97	22.43
20- 11-1/8	20.923	8- 8		6- 1-9/16	362.66	926.41	22.65
21- 1-1/2	21.124	8- 9		6- 2-1/4	369.66	953.37	22.86
21- 3-7/8	21.325	8- 10		6- 2-15/16	376.73	980.84	23.05
21- 6-1/4	21.527	8- 11		6- 3-5/8	383.90	1008.98	23.30
21- 8-3/4	21.728	9- 0		6- 4-3/8	391.10	1037.50	23.52
21- 11-1/4	21.929	9- 1		6- 5-1/16	398.38	1067.57	23.74
22- 1-5/8	22.130	9- 2		6- 5-13/16	405.72	1096.22	23.95
22- 4	22.331	9- 3		6- 6-1/2	413.17	1126.31	24.17
22- 6-3/8	22.533	9- 4		6- 7-3/16	420.62	1157.15	24.39
22- 8-7/8	22.734	9- 5		6- 7-15/16	428.15	1188.39	24.61
22- 11-1/4	22.935	9- 6		6- 8-5/8	435.76	1220.19	24.83
23- 1-5/8	23.136	9- 7		6- 9-5/16	443.44	1252.56	25.04
23- 4	23.337	9- 8		6- 10	451.17	1285.49	25.26
23- 6-1/2	23.539	9- 9		6- 10-3/4	459.02	1319.16	25.49
23- 8-7/8	23.739	9- 10		6- 11-7/16	466.85	1353.07	25.72
23- 11-3/8	23.941	9- 11		7- 0-3/16	474.83	1387.90	25.92
24- 1-3/4	24.142	10- 0		7- 0-7/8	482.84	1423.15	26.14
24- 4-1/8	24.343	10- 1		7- 1-9/16	490.91	1459.00	26.35
24- 6-1/2	24.545	10- 2		7- 2-1/4	499.09	1495.62	26.56
24- 9	24.746	10- 3		7- 3	507.30	1532.67	26.79
24- 11-3/8	24.947	10- 4		7- 3-11/16	515.57	1570.32	27.00
25- 1-3/4	25.148	10- 5		7- 4-3/8	523.92	1608.58	27.22
25- 4-1/4	25.349	10- 6		7- 5-1/8	532.32	1647.46	27.44
25- 6-5/8	25.550	10- 7		7- 5-13/16	540.80	1686.96	27.66
25- 9	25.752	10- 8		7- 6-1/2	549.38	1727.29	27.87
25- 11-1/2	25.953	10- 9		7- 7-1/4	557.99	1768.05	28.10
26- 1-7/8	26.154	10- 10		7- 7-15/16	566.67	1809.45	28.31
26- 4-3/8	26.355	10- 11		7- 8-11/16	575.41	1851.49	28.54
26- 6-5/8	26.556	11- 0		7- 9-5/16	584.22	1894.18	28.74
26- 9-1/8	26.757	11- 1		7- 10-3/16	593.10	1937.51	28.96
26- 11-1/2	26.959	11- 2		7- 10-3/4	602.09	1981.73	29.18
27- 1-7/8	27.160	11- 3		7- 11-7/16	611.10	2026.38	29.39
27- 4-3/8	27.361	11- 4		8- 0-3/16	620.18	2071.71	29.62
27- 6-3/4	27.562	11- 5		8- 0-7/8	629.33	2117.70	29.83
27- 9-1/8	27.763	11- 6		8- 1-9/16	638.54	2164.37	30.05
27- 11-5/8	27.965	11- 7		8- 2-5/16	647.86	2217.96	30.27

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Octagon Properties

D		C		B	A	Z	E
FT	IN.	FT	IN.	FT	FT ²	FT ³	FT
28-	2	28.165	11-	8	657.21	2260.00	30.49
28-	4-1/2	28.376	11-	9	666.62	2308.73	30.70
28-	6-3/4	28.568	11-	10	675.86	2356.91	30.92
28-	9-1/4	28.769	11-	11	685.65	2408.29	31.14
28-	11-5/8	28.970	12-	0	695.31	2459.36	31.36
29-	2-1/8	29.171	12-	1	704.95	2510.65	31.57
29-	4-1/2	29.373	12-	2	714.74	2563.16	31.79
29-	6-7/8	29.574	12-	3	724.56	2616.15	32.01
29-	9-1/4	29.775	12-	4	734.44	2669.85	32.23
29-	11-3/4	29.977	12-	5	744.44	2724.56	32.45
30-	2-1/8	30.178	12-	6	754.45	2779.73	32.66
30-	4-1/2	30.379	12-	7	764.54	2838.65	32.88
30-	7	30.580	12-	8	774.69	2892.30	33.10
30-	9-3/8	30.781	12-	9	784.91	2949.72	33.32
30-	11-3/4	30.982	12-	10	795.19	3007.88	33.53
31-	2-1/4	31.184	12-	11	805.60	3067.09	33.75
31-	4-5/8	31.385	13-	0	816.01	3126.79	33.99
31-	7	31.586	13-	1	826.50	3187.25	34.19
31-	9-1/2	31.787	13-	2	837.05	3248.45	34.41
31-	11-7/8	31.988	13-	3	847.67	3310.50	34.62
32-	2-1/4	32.189	13-	4	858.35	3373.30	34.84
32-	4-3/4	32.391	13-	5	869.17	3437.20	35.06
32-	7-1/8	32.592	13-	6	879.98	3501.59	35.28
32-	9-1/2	32.793	13-	7	890.87	3566.78	35.49
33-	0	32.994	13-	8	901.82	3632.74	35.71
33-	2-3/8	33.195	13-	9	912.85	3699.56	35.93
32-	4-3/4	33.397	13-	10	923.99	3767.51	36.15
33-	7-1/8	33.597	13-	11	935.15	3835.95	36.37
33-	9-5/8	33.799	14-	0	946.37	3905.21	36.58
34-	0	34.009	14-	1	958.16	3975.29	36.80
34-	2-1/2	34.201	14-	2	969.02	4046.21	37.02
34-	4-7/8	34.403	14-	3	980.50	4118.33	37.24
34-	7-1/4	34.603	14-	4	991.97	4190.81	37.45
34-	9-3/4	34.805	14-	5	1003.55	4264.39	37.70
35-	0-1/8	35.006	14-	6	1015.18	4338.74	37.89
35-	2-1/2	35.207	14-	7	1026.87	4413.94	38.10
35-	5	35.409	14-	8	1038.65	4490.11	38.33
35-	7-3/8	35.610	14-	9	1050.48	4567.05	38.54
35-	9-3/4	35.811	14-	10	1062.38	4644.86	38.76
36-	0-1/4	36.012	14-	11	1074.36	4723.65	38.98
36-	2-5/8	36.213	15-	0	1086.40	4803.23	39.20
36-	5	36.414	15-	1	1098.50	4883.69	39.41
36-	7-1/2	36.616	15-	2	1110.68	4965.15	39.63
36-	9-7/8	36.817	15-	3	1122.91	5047.41	39.85

FLUOR

VERTICAL VESSEL FOUNDATIONS

Octagon Properties

	D		C		B		A	Z	E
	FT - IN.	FT	FT - IN.		FT - IN.		FT ²	FT ³	FT
37- 0-1/4		37.018	15- 4		10- 10-1/8		1135.21	5130.58	40.07
37- 2-3/4		37.219	15- 5		10- 10-7/8		1147.59	5214.75	40.29
37- 5-1/8		37.420	15- 6		10- 11-9/16		1160.03	5299.74	40.50
37- 7-1/2		37.621	15- 7		11- 0-1/4		1172.53	5385.64	40.72
37- 9-7/8		37.822	15- 8		11- 0-15/16		1185.12	5472.58	40.94
38- 0-3/8		38.024	15- 9		11- 1-11/16		1197.75	5560.34	41.16
38- 2-3/4		38.225	15- 10		11- 2-3/8		1210.45	5649.03	41.37
38- 5-1/8		38.426	15- 11		11- 3-1/16		1223.24	5738.76	41.59
38- 7-5/8		38.627	16- 0		11- 3-13/16		1236.08	5829.34	41.81
38- 10		38.829	16- 1		11- 4-1/2		1248.98	5920.86	42.03
39- 0-3/8		39.030	16- 2		11- 5-3/16		1261.97	6013.45	42.25
39- 2-7/8		39.231	16- 3		11- 5-15/16		1275.00	6106.88	42.46
39- 5-1/4		39.432	16- 4		11- 6-5/8		1288.11	6201.28	42.68
39- 7-5/8		39.633	16- 5		11- 7-5/16		1301.30	6296.76	42.90
39- 10-1/8		39.835	16- 6		11- 8-1/16		1314.54	6393.09	43.12
40- 0-1/2		40.036	16- 7		11- 8-3/4		1327.85	6490.41	43.33

FLUOR

VERTICAL VESSEL FOUNDATIONS

Footing Thickness Using No Top Steel

(Ultimate Strength Design, $f'_c = 3000$ PSI)

The curves below are for determining the thickness of the base if no top steel is used. Figured for weight of concrete and soil acting on 1 inch strip.

Basis of Design:

$$f'_c = 3000 \text{ psi}$$

$$\phi = 0.65$$

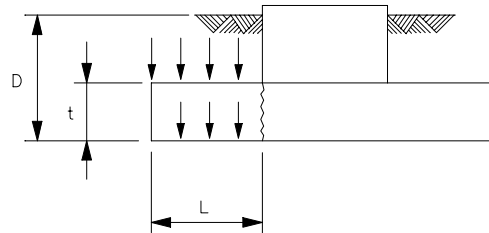
$$\text{concrete} = 150 \text{ \#/CF}$$

$$f_t = 5 \phi \sqrt{f'_c}$$

$$\text{soil} = 100 \text{ \#/CF}$$

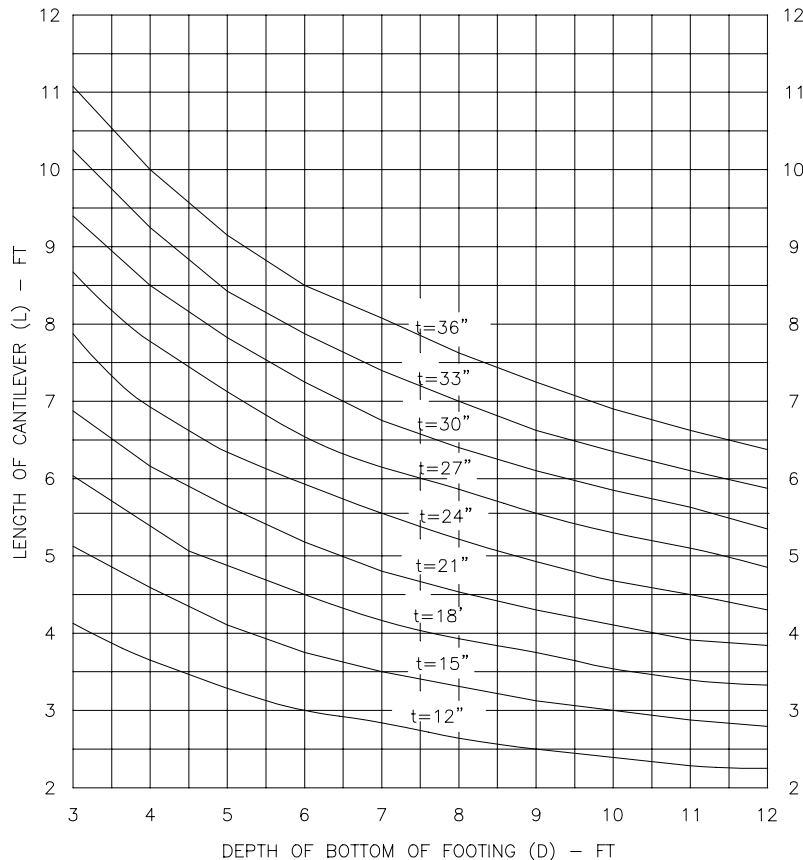
$$t = \sqrt{6Mu/f_t}$$

$$Mu = M \times 1.4$$



NOTE:

Where concrete is cast against soil, (for example, not on a seal slab), the "t" value below will be increased by 2 inches (ACI 318.1, Section 6.3.5).



FLUOR

VERTICAL VESSEL FOUNDATIONS

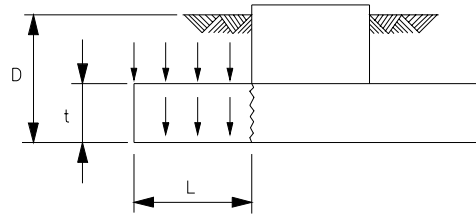
Footing Thickness Using No Top Steel

(Ultimate Strength Design, $f'_c = 4000$ PSI)

The curves below are for determining the thickness of base if no top steel is used. Figured for weight of concrete and soil acting on 1 inch strip.

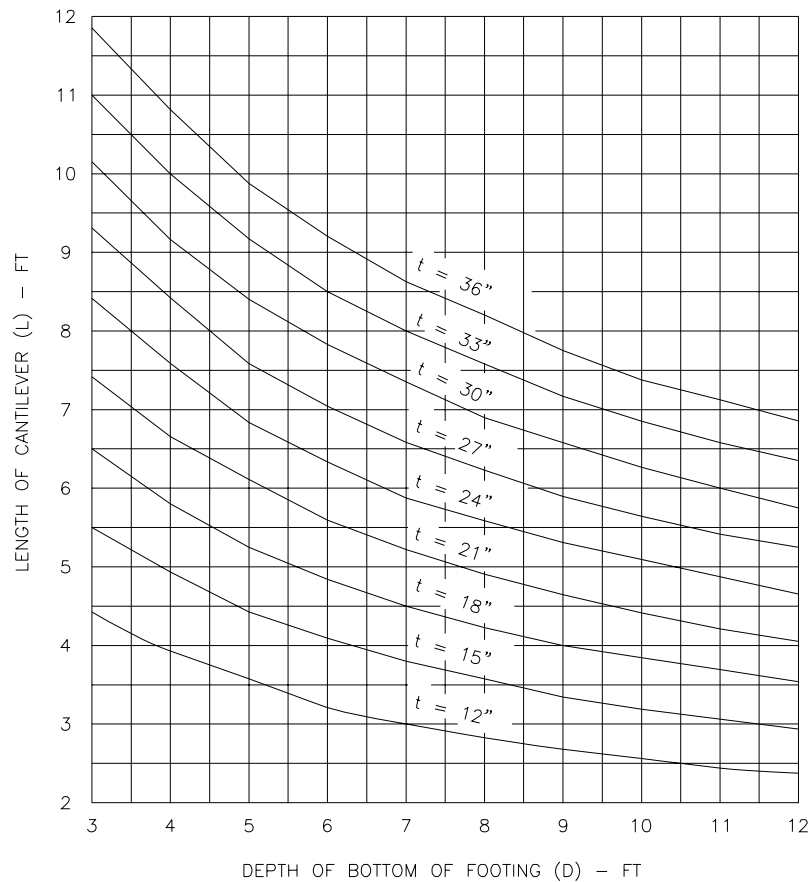
Basis of Design

$$\begin{aligned} f'_c &= 4000 \text{ psi} & \phi &= 0.65 \\ \text{concrete} &= 150 \text{ \#/CF} & f_t &= 5 \phi \sqrt{f'_c} \\ \text{soil} &= 100 \text{ \#/CF} & t &= \sqrt{6Mu/f_t} \\ Mu &= M \times 1.4 \end{aligned}$$



NOTE:

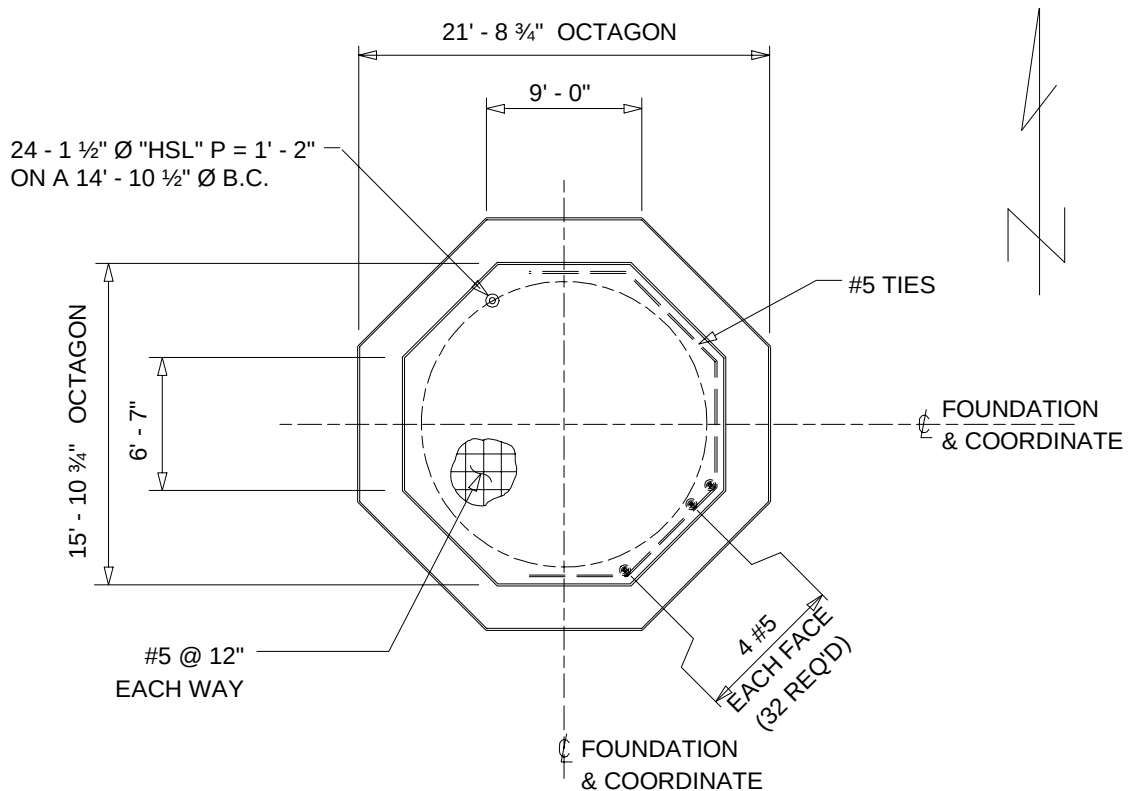
Where concrete is cast against soil, (for example, not on a seal slab), the "t" value below will be increased by 2 inches (ACI 318.1, Section 6.3.5).



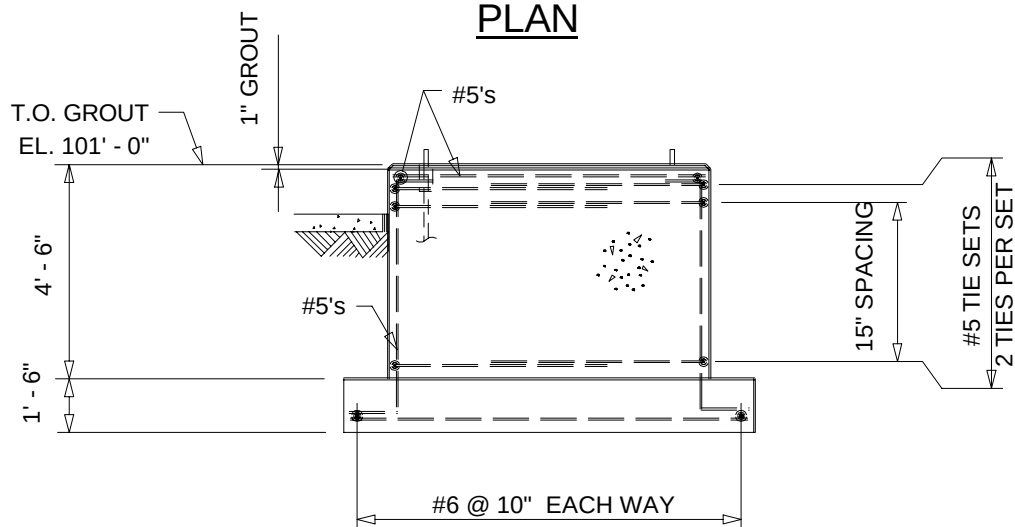
FLUOR

VERTICAL VESSEL FOUNDATIONS

Sample Design 1: Vertical Vessel Foundation



PLAN



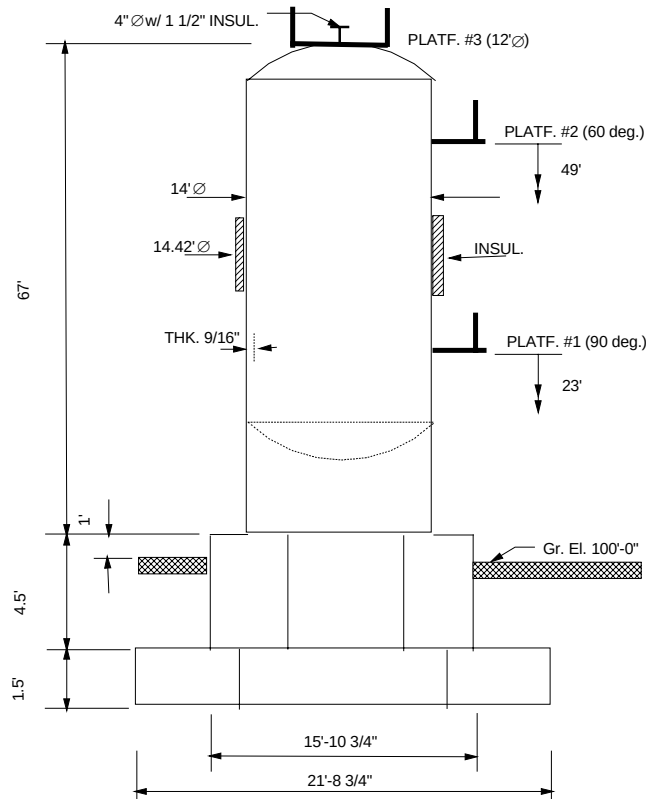
SECTION

FLUOR

VERTICAL VESSEL FOUNDATIONS

Sample Design 1: Vertical Vessel Foundation

DESIGN DATA



Wind Pressures ($q_z G_z$) per ASCE 7 (70 mph)

0-30' : 40 psf
30-50' : 50 psf
50-100' : 88 psf

From vessel drawing:

Empty Wt. = 170.3 K
Oper. Wt. = 345.2 K
Test Wt. = 624.1 K

From structural criteria spec:

Soil Bearing = 3.25 KSF (Net)
 $g = 110$ PCF
 $f'_c = 3,000$ PSI
 $f_y = 60,000$ PSI

From vessel's computer output:

$H = 44.75$ K
 $M = 1902.11$ K-FT

Anchor Bolts : 24 - 1 1/2" Ø Type MS
w/ 1'-2" projection on a 14' - 10 1/2" Ø
bolt circle.

Wind Load Verification

Note: Wind load calculations are not shown for brevity. See technical practice 000 215 1215 "Wind Load Calculations" for procedures.

Results from the vessel program are confirmed.

PEDESTAL DESIGN

$$BC + 8" = 178.5" + 8" = 186.5" \quad \text{(Equation 2)}$$

$$BC + 8(BD) = 178.5" + 8(1.5") = 190.5" \quad \leftarrow \text{Controls} = 15.875'$$

$$BC + SD + 6" = 178.5" + 4" + 6" = 188.5"$$

USE : 15' - 10 3/4" Octagon

$$\text{Area} = 209.25 \text{ FT}^2$$

(Attachment 02)

$$\text{Pedestal Weight} = 209.25 \text{ FT}^2 (4.5 \text{ FT}) (0.15 \text{ KCF}) = 141.2 \text{ K}$$

FLUOR

VERTICAL VESSEL FOUNDATIONS

Sample Design 1: Vertical Vessel Foundation

Pedestal Reinforcing

$$M_{ped} = \text{O.T.M. @ pedestal base} = 1902.11 \text{ K-FT} + 4.5 \text{ FT}(44.75 \text{ K}) = 2103.5 \text{ K-FT}$$

N = No. of bars = assume 32

$$\text{Bar Circle} = (15.90 \text{ FT pedestal}) - (\text{say } 0.5 \text{ FT}) = 15.4 \text{ FT}$$

$$W = \text{Empty Weight of Vessel} + \text{Pedestal} = 170.3 \text{ K} + 141.2 \text{ K} = 311.5 \text{ K}$$

$$F_u = [(1.3)(4M_{ped}) / (N)(\text{Bar Circle})] - (0.9W / N) \quad (\text{Equation 3b})$$
$$= [(1.3)(4)(2103.5 \text{ K-FT}) / (32)(15.4 \text{ FT})] - [(0.9)(311.5 \text{ K}) / 32] = 13.44 \text{ K}$$

$$A_s = F_u / 0.9 f_y = 13.44 \text{ K} / (0.9)(60 \text{ KSI}) = 0.24 \text{ IN}^2 \quad (\text{Equation 4})$$

USE 32 #5 Bars (As = 0.31 in²) w/ #5 ties @ 15" c/c

ANCHOR BOLT CHECK

Note: Anchor bolt calculations not shown for brevity. See technical practice 670 215 1207 "Anchor Bolt Design Criteria" for design procedures.

The anchor bolt size is adequate. Additional reinforcing is not required.

FOOTING DESIGN

Size Footing

Select a trial octagon size:

$$M_{ftg} = \text{O.T.M. @ footing base} = 1902.11 \text{ K-FT} + (6.0 \text{ FT})(44.75 \text{ K}) = 2170.6 \text{ K-FT}$$

$$SB = \text{gross soil bearing} = (3.25 \text{ KSF}) + (5 \text{ FT})(0.11 \text{ KCF}) = 3.80 \text{ KSF}$$

$$\text{Trial Diameter} = (2.6)(M_{ftg} / SB)^{1/3} = (2.6)(2170.6 \text{ K-FT} / 3.80 \text{ KSF})^{1/3} = 21.57 \text{ FT} \quad (\text{Equation 5})$$

$$\text{Try: } 21' - 8 \frac{3}{4}" \text{ octagon} \quad \text{Area} = 391.10 \text{ FT}^2 \quad (\text{Attachment 02})$$

Check required thickness without using top steel:

$$\text{Pedestal equivalent square} = (209.25 \text{ FT})^{0.5} = 14.47 \text{ FT}$$

$$L = \text{Distance from edge of footing to face of equivalent square}$$
$$= (21.73 \text{ FT} / 2) - (14.47 \text{ FT} / 2) = 3.63 \text{ FT}$$

$$D = \text{depth to bottom of footing} = 5 \text{ FT}$$

FLUOR

VERTICAL VESSEL FOUNDATIONS

Sample Design 1: Vertical Vessel Foundation

Required thickness using no top steel = 14 IN + 2 IN (no mud mat) = 16 IN
Note: This controls the footing thickness.

(Attachment 03)

Check required thickness for pedestal reinforcement embedment:

For #5 hooked bar,

$$L_{hb} = 1200 (db) / \sqrt{f'_c} \quad (\text{ACI 318-95, Section 12.5.2})$$
$$= (1200)(5/8 \text{ IN}) / \sqrt{3,000 \text{ PSI}} = 13.7 \text{ IN}$$

$$A_s \text{ req'd} / A_s \text{ prov} = 0.24 \text{ IN}^2 / 0.31 \text{ IN}^2 = .77$$

$$\text{Treq'd} = (3 \text{ IN clear}) + (2 \text{ layers})(0.75 \text{ IN bar}) + (.77)(0.7)(13.7 \text{ IN}) = 11.9 \text{ IN}$$

Try footing thickness = 18 IN

Footing Weights:

$$\text{Net weight of pedestal} = (209.25 \text{ FT}^2)[4.5 \text{ FT}(0.15 \text{ KCF}) - 3.5 \text{ FT}(0.11 \text{ KCF})] = 60.7 \text{ K}$$

$$\text{Weight of footing + soil} = (391.10 \text{ FT}^2)[1.5 \text{ FT}(0.15 \text{ KCF}) + 3.5 \text{ FT}(0.11 \text{ KCF})] = 238.6 \text{ K}$$

$$\text{Total} = (60.7 \text{ K}) + (238.6 \text{ K}) = 299.3 \text{ K}$$

$$\text{Empty WT. + FDN} = 170.3 \text{ K} + 299.3 \text{ K} = 469.6 \text{ K}$$

$$\text{Oper. WT. + FDN} = 345.2 \text{ K} + 299.3 \text{ K} = 644.5 \text{ K}$$

$$\text{Test WT. + FDN} = 624.1 \text{ K} + 299.3 \text{ K} = 923.4 \text{ K}$$

Check soil bearing and stability:

$$\text{Empty + Wind:} \quad P = 469.6 \text{ K} \quad M_{ftg} = 2170.6 \text{ K-FT}$$

$$e = M_{ftg} / P = (2170.6 \text{ K-FT}) / (469.6 \text{ K}) = 4.62 \text{ FT}$$

$$e / D = (4.62 \text{ FT}) / (21.73 \text{ FT}) = 0.212 > 0.123 \quad \therefore L = 2.95 \quad (\text{Attachment 01})$$

$$\text{Stability Ratio} = B / 2 E = 21.73 \text{ FT} / (2(4.62 \text{ FT})) = 2.35 > 1.5 \quad \therefore \text{O.K.}$$

$$\text{S.B.} = L P / A = (2.95)(469.6 \text{ K}) / (391.1 \text{ FT}^2) = 3.54 \text{ KSF} < 3.80 \text{ KSF}$$

$$\text{Operating + Wind:} \quad P = 644.5 \text{ K} \quad M_{ftg} = 2170.6 \text{ K-FT}$$

$$e = M_{ftg} / P = (2170.6 \text{ K-FT}) / (644.5 \text{ K}) = 3.37 \text{ FT}$$

$$e / D = (3.37 \text{ FT}) / (21.73 \text{ FT}) = 0.155 > 0.123 \quad \therefore L = 2.30 \quad (\text{Attachment 01})$$

FLUOR

VERTICAL VESSEL FOUNDATIONS

Sample Design 1: Vertical Vessel Foundation

$$\text{Stability Ratio} = B / 2 E = (21.73 \text{ FT}) / (2(3.37 \text{ FT})) = 3.22 > 1.5 \therefore \text{O.K.}$$

$$\text{S.B.} = L P / A = (2.30)(644.5 \text{ K}) / (391.1 \text{ FT}^2) = 3.79 \text{ KSF} < 3.80 \text{ KSF} \therefore \text{O.K. (controlling case)}$$

Test + 35 mph Wind: $P = 923.4 \text{ K}$

$$M_{\text{fig}} = (35 \text{ mph} / 70 \text{ mph})^2 (1902.11 \text{ FT-K}) = 475.5 \text{ FT-K}$$

$$e = M_{\text{fig}} / P = (475.5 \text{ FT-K}) / (923.4 \text{ K}) = 0.51 \text{ FT}$$

$$e/D = (0.51 \text{ FT}) / (21.73 \text{ FT}) = 0.023$$

$$\begin{aligned} \text{S.B.} &= P / A [1 + 8.19 (e/D)] \\ &= (923.4 \text{ K}) / (391.1 \text{ FT}^2) [1 + 8.19 (0.023)] = 2.81 \text{ KSF} < 3.80 \text{ KSF} \therefore \text{O.K.} \end{aligned} \quad (\text{Equation 7a})$$

USE : 21' - 8 3/4" Octagon

Footings Reinforcing

Check Operating + Wind (in this instance, 0.9D + 1.3W controls)

$$P_u = 0.9 (646.0 \text{ K}) = 581.4 \text{ K}$$

$$M_u = 1.3 (2170.6 \text{ K-FT}) = 2821.8 \text{ K-FT}$$

$$e = M / P = (2821.8 \text{ K-FT}) / (581.4 \text{ K}) = 4.85 \text{ FT}$$

$$e / D = (4.85 \text{ FT}) / (21.73 \text{ FT}) = 0.223 > 0.123$$

$$L = 2.95 \text{ (flat)} \quad K = 0.29 \text{ (flat)} \quad (\text{Attachment 01})$$

$$KD = .29 (21.73 \text{ FT}) = 6.3 \text{ FT}$$

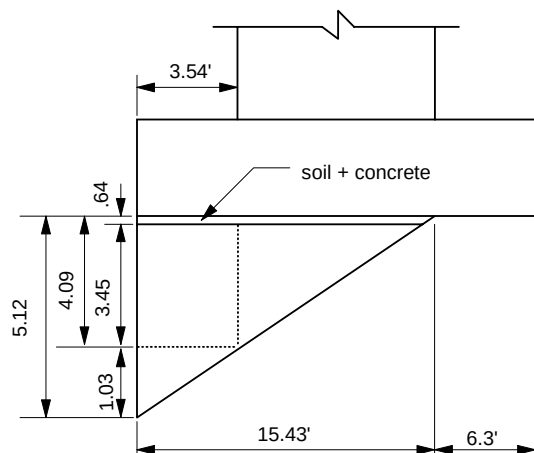
$$\text{S.B.} = L P/A = 2.95 (678.3 \text{ K}) / 391.1 \text{ FT}^2 = 5.12 \text{ KSF}$$

$$\begin{aligned} \text{S.B. @ face of equivalent square :} \\ &= 5.12 \text{ KSF (14.06 FT)} / (17.60 \text{ FT}) = 4.09 \text{ KSF} \end{aligned}$$

$$\begin{aligned} \text{soil + concrete} &= 1.05 (238.6 \text{ K}) / 391.1 \text{ FT}^2 \\ &= 0.64 \text{ KSF} \end{aligned}$$

$$\begin{aligned} V_u &= 3.45 \text{ KSF (3.54 FT)} + 1.03 \text{ KSF (3.54 FT)} / 2 \\ &= 12.21 \text{ K} + 1.82 \text{ K} = 14.03 \text{ K} \end{aligned}$$

$$\begin{aligned} M_u &= 12.21 \text{ K (3.54 FT)} / 2 + 1.82 \text{ K (3.54 FT)(2/3)} \\ &= 25.91 \text{ K-FT} \end{aligned}$$



FLUOR

VERTICAL VESSEL FOUNDATIONS

Sample Design 1: Vertical Vessel Foundation

$$d = 18" - 3" - 1.5" = 13.5"$$

$$v_u = (14.03 \text{ K})(1000) / [(12 \text{ IN})(13.5 \text{ IN})] = 86.6 \text{ PSI} < 2 \phi \sqrt{f'_c} = 93.1 \text{ PSI} \quad \therefore \text{O.K.}$$

$$F = bd^2 / 12000 = 12 \text{ IN} (13.5 \text{ IN})^2 / 12000 = 0.182$$

$$K_u = M_u / F = 25.91 \text{ K-FT} / 0.182 = 142 \quad a_u = 4.36 \quad (\text{ACI Handbook, Flexure 2.1})$$

$$A_s = M_u / (a_u d) = 25.91 \text{ K-FT} / (4.36)(13.5 \text{ IN}) = 0.44 \text{ IN}^2/\text{FT}$$

$$A_s \text{ min} = 0.0033 (12 \text{ IN})(13.5 \text{ IN}) = 0.53 \text{ IN}^2/\text{FT} \quad \leftarrow \text{Controls}$$

$$4/3 A_s = 4/3 (0.44 \text{ IN}^2/\text{FT}) = 0.59 \text{ IN}^2/\text{FT}$$

USE #6 @ 10" E.W. (bottom) $A_s = 0.53 \text{ in}^2/\text{ft}$